

SOFTWARE REQUIREMENT SPECIFICATION (SRS)

Project Title

E-Commerce Order Management Database System

1. Introduction

1.1 Purpose

The purpose of this Software Requirement Specification (SRS) document is to define the functional and non-functional requirements of the E-Commerce Order Management Database System. This document serves as a reference for database design, development, implementation, testing, and maintenance.

1.2 Project Overview

The E-Commerce Order Management Database System is designed to manage the daily operations of an online shopping platform. The system stores customer information, product details, supplier records, inventory, customer orders, payments, shipment details, and customer reviews in a centralized database.

The system improves operational efficiency, minimizes data redundancy, ensures data accuracy, and supports business decision-making through report generation.

1.3 Objectives

The main objectives of the system are:

- To provide a centralized database.
- To maintain accurate customer records.
- To manage products and inventory.
- To process customer orders efficiently.
- To maintain payment information.
- To track shipments.
- To store customer reviews.
- To generate business reports.

2. Overall Description

2.1 Product Perspective

The proposed system is a centralized database application developed for an online shopping company. It integrates customer management, product management, supplier management, inventory, orders, payments, shipments, and reviews into a single database system.

2.2 Product Features

The system provides the following features:

- Customer Registration
 - Customer Information Management
 - Category Management
 - Product Management
 - Supplier Management
 - Inventory Tracking
 - Order Processing
 - Payment Management
 - Shipment Tracking
 - Customer Reviews
 - Business Report Generation
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2.3 User Classes

Customer

- Register an account
- Place orders
- Make payments
- Track shipments
- Submit product reviews

Administrator

- Manage customers
- Manage categories
- Manage products
- Manage suppliers
- Process orders

- Manage shipments
- Generate reports

Supplier

- Supply products
- Update product availability

Warehouse Staff

- Update stock quantity
- Prepare products for shipment

Delivery Staff

- Deliver customer orders
 - Update delivery status
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3. System Requirements

3.1 Functional Requirements

Customer Module

- Register customers.
- Store customer details.
- Update customer information.
- View customer profile.

Category Module

- Add categories.
- Update categories.
- Delete categories.
- View category details.

Product Module

- Add products.
- Update product information.
- Delete products.
- Maintain stock quantity.

Supplier Module

- Store supplier information.
- Update supplier details.
- View supplier records.

Order Module

- Place orders.
- Update order status.
- Calculate total amount.
- View order history.

Order Details Module

- Store ordered products.
- Record quantity.
- Record unit price.

Payment Module

- Store payment information.
- Update payment status.
- Maintain payment history.

Shipment Module

- Store shipment details.
- Update delivery status.
- Track shipments.

Review Module

- Record customer ratings.
- Store comments.
- Display reviews.

Reporting Module

The system should generate reports for:

- Customers
 - Products
 - Inventory
 - Orders
 - Payments
 - Shipments
 - Reviews
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4. Non-Functional Requirements

Performance

- Fast query execution.
 - Quick order processing.
 - Support multiple users simultaneously.
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Security

- Secure customer information.
 - Restrict unauthorized access.
 - Protect payment information.
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Reliability

- Ensure accurate data.
 - Prevent data loss.
 - Maintain database consistency.
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Availability

- System available 24×7.
 - Minimum downtime.
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Usability

- User-friendly interface.
 - Easy navigation.
 - Simple data entry.
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Scalability

- Support increasing customers.
- Handle growing product database.
- Manage increasing order volume.

Maintainability

- Easy software maintenance.
 - Easy database modification.
 - Support future enhancements.
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5. Core Entities

The database includes the following entities:

Entity	Description
Customer	Stores customer information
Category	Stores product categories
Product	Stores product details
Supplier	Stores supplier information
Order	Stores customer orders
Order Details	Stores products in each order
Payment	Stores payment records
Shipment	Stores delivery information
Review	Stores customer feedback

6. Assumptions

- Customers must register before placing orders.
 - Products belong to one category.
 - Payments are completed before shipment.
 - Reviews are submitted by registered customers.
 - Stock is updated after every successful order.
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7. Constraints

- Only the given entities and attributes are used.
- The database is centralized.
- Every order belongs to one customer.
- Every payment is linked to one order.
- Every shipment belongs to one order.
- Each review is linked to one customer and one product.

8. Project Scope

The system includes:

- Customer Management
- Product Management
- Category Management
- Supplier Management
- Inventory Management
- Order Management
- Payment Management
- Shipment Tracking
- Review Management
- Business Report Generation

The system does not include advanced features such as AI recommendations, online chat support, or promotional campaigns.

9. Expected Outcomes

After implementation, the system will:

- Improve customer satisfaction.
 - Reduce data redundancy.
 - Increase operational efficiency.
 - Maintain accurate records.
 - Support business growth.
 - Provide reliable reports for decision-making.
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10. Future Enhancements

The system can be enhanced by adding:

- Online shopping cart
- Discount and coupon management
- Multiple payment gateways
- Email and SMS notifications
- Mobile application support
- AI-based product recommendations
- Sales analytics dashboard

11. Conclusion

The E-Commerce Order Management Database System provides a centralized and efficient solution for managing the operations of an online shopping business. The Software Requirement Specification defines all essential requirements for the successful design and implementation of the database system. It serves as the foundation for future stages such as ER modeling, normalization, SQL implementation, and report generation.